Ram Calculus for Coulomb Densities without Kohn–Sham Ensemble Representation: Weighted Domains, Certified Gap Tests, and a Larger Excluded Potential Class

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M561

October 8, 2026

Abstract

The source manuscript constructs a three-electron, two-centre Coulomb density and asserts that no real, spin-independent scalar potential in $L^{3/2}(\mathbb{R}^3) + L^{\infty}(\mathbb{R}^3)$ represents it as a noninteracting ground-state ensemble. Its essential mechanism couples a nodal-plane density tail to a weighted spectral gap, a Pauli-saturation argument, and an incompatible exterior decay estimate. We separate the source's substantial Coulomb channel analysis from its fixed-density obstruction, and reformulate the latter using the domain-aware and physical-metric principles of Ram Calculus. We prove a quantitative angle-residual identity, a metric-correct finite/complement gap criterion, density and occupation defect bounds, and a conditional extension of the obstruction to locally vanishing form-small remainders beyond global $L^{3/2}$. A sparse positive-spike example proves that this potential class is strictly larger. These results improve the organization and reach of the fixed-density argument; they neither numerically instantiate the source's very large nuclear charge nor independently certify all of its channel asymptotics. Current one-dimensional representability and general uniqueness results are distinguished from the three-dimensional existence question. Contribution assessment: C3/5, a notable conditional extension and explicit error and gap certificates without independent certification of the complete Coulomb construction.

Contents

1 Question, scope, and evidence **2**
1.1 The precise question . . . 2
1.2 What has been examined . . . 2
1.3 Relevant Ram Calculus material . . . 3

2 Coulomb model and the source's construction **3**
2.1 Scaling, spaces, and symmetry . . . 3
2.2 Channel analysis and its nontrivial dependencies . . . 4
2.3 Tail information actually used downstream . . . 5

3 Angle minimization and exact scalar inversion **5**

4 Ram weighted realization with an explicit domain **6**

5 Gap derivation and quantitative certification **7**
5.1 Metric-correct finite and complementary spaces 8

6 Pauli saturation and arbitrary ensembles **8**

7 The fixed-density nodal-plane obstruction **10**

8 A strictly larger excluded potential class **11**

9 Charge selection and what remains nonnumerical **13**

10 Functional consequences and limits of differentiability claims **14**

11 Current literature and noncontradictions **14**

12 Implementation and reproducibility plan **15**

13 Advantages, limitations, and future work **15**
13.1 Advantages over an unstructured reformulation 15
13.2 Costs and boundaries 15
13.3 Research directions 16

14 Assessment and conclusion **16**

1 Question, scope, and evidence

1.1 The precise question

For a density $n \geq 0$ of mass three on $\mathbb{R}^3$, does there exist a real, local, spin-independent scalar potential v such that n is the density of a ground-state ensemble of three noninteracting spin-1/2 fermions? The ensemble may be mixed, have infinite rank, and involve degeneracies and complex spinors. No assumption of a unique Kohn–Sham determinant is appropriate. The potential class, form domain, and meaning of ground state are indispensable parts of the question.

The source, identified as M561 in the fixed 722-manuscript catalogue, is *A Coulomb ground-state density without Kohn–Sham ensemble representation*, September 25, 2026 [1]. Its original potential class is

$$\mathcal{V} = L^{3/2}(\mathbb{R}^3;\mathbb{R}) + L^{\infty}(\mathbb{R}^3;\mathbb{R}). \tag{1}$$

Real potentials in this class are infinitesimally form bounded relative to $-\Delta$. They define closed, lower-semibounded one-particle forms on $H^1(\mathbb{R}^3)$. The obstruction concerns exact ground-state ensemble representation in this class, not the existence of useful approximate density-functional calculations.

1.2 What has been examined

The local source's main text, all seven substantive sections, and references were read. The difficult parts are the moving ionic cutoff and channel expansion, the uniform crossover estimates, the minimizer-uniform weighted gap, and the selection of a finite integer charge through a nonnumerical supremum. The section-level account below records these dependencies rather than replacing them

with an unsupported claim of complete proof verification. Algebraic and finite-dimensional tests of the present report are not tests of the three-electron continuum ground state.

The source snapshot is commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. All eleven local text files in its subtree match their Git blobs in that snapshot and in the captured October 8 comparison commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. This establishes dated byte identity, not proof validity or a fresh live-branch status. The public repository now has a changing manuscript count and mixed verification status; the historical catalogue identifier M561 is retained. Neither repository inclusion nor an absence of a searched-for formalization file establishes mathematical validity or invalidity.

1.3 Relevant Ram Calculus material

The relevant supporting material is specific, not the entire private corpus by assertion. The physical-metric chapter of the Ram Calculus books [2] supplies the distinction between synthesis, Gram matrix, weak stiffness, and physical residual. The state/domain chapter [3] requires transporting domains and changing the Hilbert space when a weight has an unbounded inverse. The spectral-limits chapter [4] explains why a finite coefficient matrix is not by itself a continuum spectral certificate. The PDE-first and form-domain discussion in TR8 [5] supports preserving boundary, trace, and operator conditions under reformulation. These are organizational principles and standard operator mechanisms with explicit proofs here; no private theorem is invoked as a black box to establish the source's channel expansion.

2 Coulomb model and the source's construction

2.1 Scaling, spaces, and symmetry

Let two equal nuclei of charge Z lie at $\pm(D/Z)e_z$. The physical Hamiltonian acts on

$$\bigwedge^3 L^2(\mathbb{R}^3;\mathbb{C}^2)$$

with kinetic energy $-\frac{1}{2}\Delta$ per electron and Coulomb repulsion. Under $x = Zr$, it becomes $Z^2 H_{3,\lambda}$, where $\lambda = Z^{-1}$ and

$$H_{m,\lambda} = \sum_{i=1}^{m}\left[-\frac{1}{2}\Delta_i - \frac{1}{|x_i - De_z|} - \frac{1}{|x_i + De_z|}\right] + \lambda\sum_{i<j}\frac{1}{|x_i - x_j|}. \tag{2}$$

The form domain is the antisymmetric H^1 space; the Coulomb operator domain is the corresponding H^2 space. Spin and reflection sectors are specified before perturbation. Scaling sends a scaled density to $Z^3 n(Zr)$ and a physical potential to $Z^{-2}v(x/Z)$. The $L^{3/2}$ norm of the latter is unchanged; boundedness is preserved with the appropriate factor. Thus the representability question respects this dilation.

For the one-electron two-centre operator, let $b > 0$ be its even ground orbital and e its odd second orbital, positive for $z > 0$, with

$$\epsilon_b = -\gamma^2/2, \qquad \epsilon_e = -k_0^2/2, \qquad d_* = \epsilon_e - \epsilon_b > 0.$$

The source uses an IMS localization to separate these two low orbitals from the rest. For $D \geq 100$, the displayed lower and upper bounds are, respectively,

$$-\tfrac{1}{2} - D^{-1} - 20D^{-2}, \qquad -\tfrac{1}{2} + 40D^{-2},$$

and the third level is bounded below by $-\frac{1}{8} - D^{-1} - 20D^{-2}$. The resulting higher-level separation exceeds 0.2 in the stated large-D regime, whereas $d_* < 100/D$ is small. A small tunnelling gap must not be confused with a uniform positive constant independent of D.

At $\lambda = 0$, the three-electron reference determinant occupies $b\uparrow, b\downarrow, e\uparrow$; the ionic state occupies $b\uparrow, b\downarrow$. The source follows a simple analytic branch in the $S_z = 1/2$ sector and separately compares sectors to identify an absolute ground state. Perturbation simplicity in one sector alone would not prove this last assertion.

2.2 Channel analysis and its nontrivial dependencies

Write E_2, E_3 for the relevant two-electron ionic and three-electron energies and

$$k = \sqrt{2(E_2 - E_3)}, \qquad A(r) = e^{-kr} r^{\beta}, \qquad \beta = (2 - 2\lambda)/k - 1. \tag{3}$$

The Hilbert-valued extraction $F_{\mathrm{true}}(x) = \sqrt{3}\Psi(x, \cdot)$ has density $n(x) = \|F_{\mathrm{true}}(x)\|^2$. With P the ionic ground-state projection, $Q = I - P$, and $B = H_{2,\lambda} - E_2$, the channel calculation needs a strictly positive lower bound for B on Q.

A moving cutoff separates bounded ionic coordinates from the distant electron. The cutoff channel F obeys

$$\left(-\tfrac{1}{2}\Delta_x + \tfrac{k^2}{2} + B + u + \lambda W\right) F = R. \tag{4}$$

The source controls R and its required derivatives by a faster exponential than the relevant one-electron tail. Multipole expressions for W are applied to localized ionic vectors with moment control; they are not global operator-norm expansions on an unrestricted ionic Hilbert space. Differentiation must respect Coulomb collision sets, which is why the source uses parallel-cluster derivatives and cutoff geometry.

The leading scalar component has an expansion

$$PF = \phi A(r)\{a_0(\omega) + a_1(\omega) r^{-1} + a_2(\omega) r^{-2} + E(r, \omega)\}. \tag{5}$$

Here a_0 is odd in ω_z and a_0/ω_z is positively bounded below. The source gives $E = O(r^{-3})$, $\partial_r E = O(r^{-4})$, and $\partial_r^2 E = O(r^{-4})$, with the needed angular and parity control. In particular the second derivative must not silently be strengthened to $O(r^{-5})$.

Set $K_j = -\frac{1}{2}\Delta_{S^2} - \frac{1}{2}(\beta - j)(\beta - j + 1)$ and let c_m denote the scalar multipole coefficients. The first recurrence is

$$a_1 = k^{-1}(K_0 + c_2)a_0, \qquad a_2 = (2k)^{-1}\{(K_1 + c_2)a_1 + c_3 a_0\}, \qquad c_2 = 0.$$

The complementary component retains a faster orbital explicitly:

$$QF = bG + \lambda A(r)(T_2 r^{-2} + T_3 r^{-3} + T_4 r^{-4}) + E_Q, \quad E_Q = O(\lambda A r^{-5}). \tag{6}$$

The source's recurrences include

$$B_Q T_2 = S_2, \quad B_Q T_3 = S_3 + 2kT_2, \quad B_Q T_4 = S_4 + 3kT_3 - (K_2 + C_2)T_2.$$

On the equator, $T_2 = T_3 = 0$ but

$$T_4|_{\omega_z = 0} = -a_0'(0) B_Q^{-2} Q Y_z \phi \neq 0, \qquad Y_z = y_{1z} + y_{2z}. \tag{7}$$

Oddness and the nonzero multiplier Y_z are essential. A formal scalar leading term that vanishes on the plane would miss this slower complementary-channel contribution.

The explicit bG term in (6) is needed in the crossover between small coupling and large radius. Spin separation supplies, for the pertinent up-spin correction S,

$$(1-2^{-1/2})(2b^2+\|S\|^2) \le \|bG+S\|^2 \le (1+2^{-1/2})(2b^2+\|S\|^2). \tag{8}$$

This prevents cancellation and gives a uniform lower scale involving $A|\omega_z|$, b, and $\lambda A r^{-4}$. Taking $r\to\infty$ before keeping track of $\lambda\to 0$ would lose exactly the estimate later used for all folded minimizers.

2.3 Tail information actually used downstream

For a fixed admissible $\lambda>0$, the source derives a plane lower bound

$$\sqrt{n(p,0)} \ge c_\lambda e^{-kp} p^{\beta-4} \quad (p\ge p_0), \tag{9}$$

where p is radial distance in the median plane. It also derives translated density limits. If $y_j = p_j\omega_j$ lies in that plane, $p_j\to\infty$, $\omega_j\to\omega$ horizontally, a positive normalization c_j gives

$$n(y_j+x)/c_j^2 \longrightarrow a^2 z^2 e^{-2k\omega\cdot x}, \qquad a>0. \tag{10}$$

For escaping centres whose heights tend to infinity in magnitude,

$$n(y_j+x)/n(y_j) \longrightarrow e^{-2k\omega\cdot x} \tag{11}$$

along a subsequence of directions. Local uniform convergence, rather than pointwise convergence on rays alone, is important. With positive local limiting density, the divergence-form angle equation permits compactness without convergence of density derivatives.

3 Angle minimization and exact scalar inversion

Let n have mass three, be even in z, be locally strictly positive, and have the regularity established in the source. Minimize

$$\int n|\nabla\theta|^2, \qquad \theta(x',-z)=-\theta(x',z), \qquad \int n\sin^2\theta = 1. \tag{12}$$

Folding allows $0\le\theta\le\pi/2$ above the plane. The density's integrability and boundedness of the folded angle give tightness for the nonlinear constraint; a merely local weak limit without tightness would not suffice. The source obtains a nonzero constraint derivative, a positive multiplier d, and

$$\operatorname{div}(n\nabla\theta) = -dn\sin(2\theta). \tag{13}$$

The strong inequalities $0<\theta<\pi/2$ hold for $z>0$. Axisymmetry of every minimizer is not assumed. The source establishes local $W^{2,p}$ regularity for $p<6$ and suitable $C^{1,\alpha}$ regularity.

Put $s=\sqrt{n}$ and

$$g = s\cos\theta, \quad h = s\sin\theta, \quad V = \frac{\Delta s}{2s} - \frac{|\nabla\theta|^2}{2} + d\sin^2\theta. \tag{14}$$

Then $g>0$, g is even, h is odd and positive above the plane,

$$\|g\|_2^2 = 2, \qquad \|h\|_2^2 = 1, \qquad g^2+h^2 = n, \qquad H_V g = 0, \quad H_V h = dh.$$

The equations are initially distributional and $V\in L^2_{\mathrm{loc}}$. They do *not* initially assert that $V\in\mathcal{V}$ or that its naive global Schrödinger form is closed on H^1.

Proposition 3.1 (Angle defect measured in physical density)**.** *For regular $s > 0$ and an approximate angle, define*

$$r = \operatorname{div}(n\nabla\theta) + dn \sin 2\theta$$

and use (14) to define g, h, V. Wherever these expressions define functions, the exact identities are

$$H_V g = \frac{\sin\theta}{2s} r, \qquad (H_V - d)h = -\frac{\cos\theta}{2s} r. \tag{15}$$

Consequently, if $r/s \in L^2$,

$$\|H_V g\|_2^2 + \|(H_V - d)h\|_2^2 = \frac{1}{4}\int \frac{|r|^2}{n}.$$

Proof. Expand $\Delta(s\cos\theta)$ and $\Delta(s\sin\theta)$. The Δs and $|\nabla\theta|^2$ terms cancel against V. Since $\operatorname{div}(s^2\nabla\theta) = 2s\nabla s \cdot \nabla\theta + s^2\Delta\theta$, the remaining terms are precisely (15). Squaring and adding uses $\sin^2\theta + \cos^2\theta = 1$. □

An unweighted residual for (13) can be small in a tail where n is tiny and yet produce a large orbital residual. The physical density weight is therefore a substantive part of a numerical stopping criterion. Dividing an approximate orbital equation by g is likewise poorly conditioned where g is small; the two-orbital identity avoids that unnecessary division.

4 Ram weighted realization with an explicit domain

Let $w = g^2$. Define the weighted space and closed gradient form by

$$\mathcal{H}_w = L^2(w\,\mathrm{d}x), \quad H^1(w) = \{\xi \in H^1_{\mathrm{loc}} : \int w(|\xi|^2 + |\nabla\xi|^2) < \infty\}, \quad \mathcal{E}_w[\xi] = \tfrac{1}{2}\int w|\nabla\xi|^2.$$

Local upper and positive lower bounds on w give completeness by local Sobolev convergence and Fatou's lemma. Constants belong to this space because $\int w = 2$. Standard cutoffs and local smoothing supply a form core. Let A_w denote the associated nonnegative self-adjoint operator, with its form-defined domain. Formally,

$$A_w\xi = -\frac{1}{2w}\operatorname{div}(w\nabla\xi).$$

The synthesis map $U\xi = g\xi$ is a unitary map from $\mathcal{H}_w$ to ordinary L^2. Thus UA_wU^{-1} is rigorously defined with domain $U\operatorname{Dom}(A_w)$. This is not a bounded similarity using g and g^{-1} as operators on the *same* unweighted space.

For compactly supported smooth f, integration by parts using $H_V g = 0$ gives

$$\frac{1}{2}\int |\nabla f|^2 + \int V|f|^2 = \mathcal{E}_w[f/g]. \tag{16}$$

The source extends this identity or inequality to the necessary potential-integrable states. A global formal potential need not be admissible for the weighted realization to exist. This separation between the transformed closed form and an asserted ordinary potential class is central to the argument, not a technical inconvenience to be suppressed.

5 Gap derivation and quantitative certification

The source needs

$$\mathcal{E}_w[\xi] \geq d\left(\int w|\xi|^2 - \frac{|\int w\xi|^2}{\int w}\right). \tag{17}$$

On odd functions, the positive upper-half-space solution h/g of the weighted eigenvalue equation gives the Dirichlet ground-state transform at energy d. The zero trace on the median plane is essential. On even functions, local closeness to $2b^2$ is combined with an exterior inequality. Local perturbation alone would not prevent low-energy mass from escaping to infinity.

The source obtains the exterior estimate through a vertical strip comparison for $\delta = \pi/2 - \theta$, a one-dimensional auxiliary function U, and a tangential outward drift. The displayed constant

$$\frac{2}{\pi^2}\left(\frac{1}{4} - \frac{1}{100} - \frac{\pi^2}{400}\right) > 0.043$$

permits the weaker exterior coefficient 0.03. Uniform bounds on derivatives of U, despite exponentially small tunnelling, are part of the source proof. They cannot be replaced by a single test at $\lambda = 0$.

Proposition 5.1 (Local metric comparison plus exterior confinement)**.** *Let w, w_0 be even positive integrable weights. Suppose w_0 has an even mean-zero gap $\Delta > 0$. Choose real, even, Lipschitz cutoffs $0 \leq \chi, \eta \leq 1$ with $\chi^2 + \eta^2 = 1$, $\chi \not\equiv 0$, such that χ is compactly supported and η is supported in the region where*

$$\mathcal{E}_w[\eta f] \geq c_{\mathrm{ext}}\|\eta f\|_w^2$$

holds for even f. Assume $0 < a \leq b$, $aw_0 \leq w \leq bw_0$ on the support of χ, and

$$\tfrac{1}{2}(|\nabla\chi|^2 + |\nabla\eta|^2) \leq \tau.$$

Then on even functions,

$$\mathcal{E}_w[\xi] \geq (\min\{a\Delta/b, c_{\mathrm{ext}}\} - \tau)\,\mathrm{Var}_w(\xi). \tag{18}$$

Proof. Set $c = (\int w_0\chi\xi)/(\int w_0\chi)$ and $f = \xi - c$. Thus χf has exactly zero w_0 mean, including for complex ξ. On the inner piece,

$$\mathcal{E}_w[\chi f] \geq a\mathcal{E}_{w_0}[\chi f] \geq a\Delta\|\chi f\|_{w_0}^2 \geq (a\Delta/b)\|\chi f\|_w^2.$$

The weighted IMS identity subtracts at most $\tau\|f\|_w^2$ from the sum of the two localized energies. Their squared norms add to $\|f\|_w^2$. Finally $\|\xi - c\|_w^2 \geq \mathrm{Var}_w(\xi)$. If the displayed coefficient is negative, the claimed lower bound follows instead from nonnegativity; for a useful certificate it is required positive. □

The mean choice is important: subtracting the w mean instead would not make the localized w_0 mean vanish. Combining (18) with the odd bound certifies (17) whenever the even coefficient is at least d.

5.1 Metric-correct finite and complementary spaces

Take linearly independent even mean-zero trial functions $\psi_1, \dots, \psi_m \in \mathrm{Dom}(A_w)$ and define

$$G_{ij} = \langle \psi_i, \psi_j \rangle_w, \quad K_{ij} = \mathcal{E}_w(\psi_i, \psi_j), \quad L_{ij} = \langle A_w\psi_i, A_w\psi_j \rangle_w.$$

Here $G > 0$ is the physical metric. The coefficient operator is $G^{-1}K$, while the Hermitian matrix for ordinary Euclidean eigenvalue calculations is $G^{-1/2}KG^{-1/2}$. They are not interchangeable with K alone.

Let P be the physical orthogonal projection onto the trial space in the even mean-zero Hilbert space. Assume a proved complementary form lower bound β on $(I-P)\mathcal{H}_w$, and let

$$\alpha = \lambda_{\min}(G^{-1/2}KG^{-1/2}), \qquad \eta^2 = \lambda_{\max}\{G^{-1/2}(L - KG^{-1}K)G^{-1/2}\}. \tag{19}$$

The residual Gram matrix in (19) uses $A_w\psi_i$ in the full physical space. A projected residual would incorrectly set this coupling to zero.

Theorem 5.2 (Finite/complement gap certificate)**.** *Under these assumptions, the even mean-zero form is bounded below by*

$$\lambda_- = \frac{\alpha + \beta - \sqrt{(\alpha-\beta)^2 + 4\eta^2}}{2}.$$

In particular a lower bound d is certified if

$$\alpha \geq d, \qquad \beta \geq d, \qquad \eta^2 \leq (\alpha - d)(\beta - d). \tag{20}$$

Proof. For $f = p + q$ in the form domain, $p = Pf$ lies in the operator domain. Hence the cross form equals $\langle (I-P)A_w p, q\rangle_w$ and is bounded in modulus by $\eta\|p\|_w\|q\|_w$. The full form is at least

$$\alpha\|p\|_w^2 - 2\eta\|p\|_w\|q\|_w + \beta\|q\|_w^2.$$

The smaller eigenvalue of this real 2×2 matrix gives the assertion. Subtracting dI and applying the two-dimensional positivity criterion gives (20). Projection of $A_w\psi_i$ onto the trial space has Gram contribution $KG^{-1}K$, proving the residual formula. □

A finite Ritz eigenvalue alone is an upper bound for an appropriate variational eigenvalue, not the required lower gap certificate. The hard complement estimate remains necessary. Proposition 5.1, analytic symmetry decomposition, or a separately proved coercive estimate can provide it; a matrix calculation cannot manufacture it.

6 Pauli saturation and arbitrary ensembles

Let $\widehat{g} = g/\sqrt{2}$. The gap and ground-state transform imply

$$q_V[f] \geq d\{\|f\|_2^2 - |\langle \widehat{g}, f\rangle|^2\}, \quad f \in \mathcal{E}_V := \{f \in H^1 : \int |V||f|^2 < \infty\}. \tag{21}$$

Locally, $V \in L^2$ and $H^1 \subset L^4$ permit form approximation; global cutoff limits require the displayed absolute integrability. In the source this is controlled at density level by

$$\int n|V| \leq \tfrac{1}{4}\|\Delta n\|_1 + \tfrac{1}{2}\|\nabla\sqrt{n}\|_2^2 + \tfrac{1}{2}\int n|\nabla\theta|^2 + d\int n < \infty.$$

The factor in the last term can be sharpened using the constraint, but finiteness is what is needed.

For any three-particle ensemble, its one-particle density matrix obeys

$$0 \le \gamma \le I, \qquad \operatorname{Tr}\gamma = 3.$$

The spin-lifted projection $P_0 = |\widehat{g}\rangle\langle\widehat{g}| \otimes I_{\mathbb{C}^2}$ has rank two, so $\operatorname{Tr}\gamma P_0 \le 2$. To form the next energy expression, additionally require $T(\gamma) < \infty$ and $\int |V| n_\gamma < \infty$. Then every occupied natural spinor has the needed form integrability, and summing (21) gives

$$T(\gamma) + \int V n_\gamma \ge d(3 - \operatorname{Tr}\gamma P_0) \ge d. \tag{22}$$

For competitors of density n, absolute potential integrability is already established; those with infinite kinetic energy cannot lower the constrained-search value. The determinant $\Phi = \widehat{g}\uparrow \wedge \widehat{g}\downarrow \wedge h\uparrow$ attains equality and has density n. Thus the ensemble constrained-search kinetic functional satisfies

$$T_s[n] = d - \int V n. \tag{23}$$

Complex orbitals, spin coherence, degeneracy and infinite-rank ensembles are all covered by the density-matrix inequalities; a proof restricted to three real orbitals would not suffice.

Suppose now an admissible v represents n. The same-density variational principle and (23) force Φ itself to be a noninteracting ground state. Vary its uniquely occupied down-spin orbital against arbitrary H^1 test functions. Orthogonality to the up-spin sector causes no restriction, and the Euler equation yields

$$(-\Delta/2 + v)g = e_g g.$$

Comparing with $H_V g = 0$ and using local positivity of g proves $v = V + e_g$ almost everywhere. Representation therefore reduces to admissibility of the inverted potential, up to a constant; it is not inferred from a general uniqueness theorem alone.

Proposition 6.1 (Quantitative density and occupation defects)**.** *Suppose a certified approximate lower bound is*

$$q_V[f] \ge d(\|f\|^2 - |\langle\widehat{g}, f\rangle|^2) - \varepsilon\|f\|^2.$$

For a three-particle γ with density m and $\int |V| m < \infty$,

$$T(\gamma) - T_s[n] \ge d(2 - \operatorname{Tr}\gamma P_0) - 3\varepsilon - \int V(m - n), \tag{24}$$

provided (23) *is exact for the reference density.*

Proof. Sum the approximate form bound, use $\operatorname{Tr}\gamma = 3$, and subtract (23). All potential pairings exist by the assumptions. □

The density penalty is bounded by $\int |V||m - n|$, not by a bare L^1 error when V is unbounded. If, under an additional proved spectral hypothesis, states outside the ground and d eigenspaces cost at least $d + \sigma$, one obtains the further nonnegative term $\sigma \operatorname{Tr}\gamma(I - P_0 - P_d)$. No such numerical σ is asserted here for the constructed molecule.

7 The fixed-density nodal-plane obstruction

The remainder of the argument can be isolated from the Coulomb construction. The fixed-density package used here has the following explicit content.

(D1) n is even in z, locally strictly positive and continuous, of mass three, with $\sqrt{n} \in H^1$, $\Delta n \in L^1$, and sufficient local regularity for the equations below. The odd angle obeys (13), its constraint, and $0 < \theta < \pi/2$ above the plane. The resulting g, h are in H^1, $V \in L^2_{\mathrm{loc}}$, $H_V g = 0$, $H_V h = dh$, and $\int |V| n < \infty$.

(D2) $d > 0$, $k > 0$, the full weighted gap (17) holds, and the form extension (21) is valid on $\mathcal{E}_V$.

(D3) Every escaping horizontal sequence has, after selecting its direction and a subsequence, the locally uniform normalized density limit (10) for some $a > 0$. Every escaping sequence with $|z_j| \to \infty$ has the positive locally uniform limit (11), after extracting a direction. These convergences in particular give uniform local L^2 bounds for the normalized orbitals. The weak divergence-form angle equation is retained under translation.

(D4) There are $c_\lambda, p_0 > 0$ and a finite real β for which (9) holds for all plane directions and $p \geq p_0$.

These are inherited analytic hypotheses, not newly established Coulomb facts in the present report. In particular (D3) is stronger than a pointwise tail on each fixed ray. The needed positive lower bound on θ at heights at least one will be derived under the contradictory admissibility assumption; it is not an output of the earlier strip estimate, which bounds $\pi/2 - \theta$ in the opposite direction.

If $V = V_c + V_b$ with $V_c \in L^{3/2}$ and $V_b \in L^\infty$, any escaping translates have $V_c(y_j + \cdot) \to 0$ in local $L^{3/2}$ and a weak-* subsequential bounded limit W for V_b. Caccioppoli bounds and Rellich compactness give local strong L^2 limits of the normalized g, h. At a plane-centred limit,

$$0 \leq g_\infty(x) \leq a|z|e^{-k\omega\cdot x}, \qquad (-\Delta/2 + W)g_\infty = 0.$$

Local elliptic regularity and the nonnegative-solution Harnack principle force $g_\infty = 0$, since it vanishes at an interior plane point. The sign of h fixes

$$h_\infty = aze^{-k\omega\cdot x}, \qquad W = d + k^2/2 =: c_\infty.$$

Here ω is horizontal, so $\Delta(ze^{-k\omega\cdot x}) = k^2 ze^{-k\omega\cdot x}$. This is the specific place where the nodal plane identifies the constant at infinity.

Before taking bulk limits, derive the uniform angle separation. On compact subsets of $z > 0$, the plane-normalized densities have uniform positive lower bounds and finite upper bounds. The bounded translated angles satisfy the weak equation (13) with these scalar elliptic coefficients and bounded right side. Caccioppoli and interior Hölder estimates give local H^1 bounds and locally uniform subsequential angle convergence. Since $g_j = \sqrt{n_j}\cos\theta_j$ and $g_\infty = 0$, every such limit is $\pi/2$. Hence $\theta(x', 1) \to \pi/2$ as $|x'| \to \infty$. Strict positivity and continuity on the remaining compact set give $\inf_{x'} \theta(x', 1) > 0$.

Choose $m > 0$ below this infimum. On $z > 1$, $-\operatorname{div}(n\nabla\theta) = dn \sin 2\theta \geq 0$. Test with a compact cutoff squared times $(m - \theta)_+$. Its boundary trace vanishes, and the resulting gradient estimate tends to zero as the cutoff radius grows because $n \in L^1$ and the truncated angle is bounded. Thus $(m - \theta)_+ = 0$ on this half-space. This proves $\theta \geq m$ for $z \geq 1$ without using bulk limits.

For bulk translates, the same divergence-form compactness, using their positive locally uniform density limits, gives a subsequential angle defect w satisfying

$$(\Delta - 2k\omega \cdot \nabla)w = d \sin 2w, \qquad 0 \leq w \leq \pi/2 - m.$$

On this interval, $d \sin 2w \geq c_0 w$ for some $c_0 > 0$. A bounded nonnegative solution is zero: maximize $w - \epsilon\sqrt{1+|x|^2}$, use the bounded gradient and Laplacian of the penalty, then let $\epsilon \downarrow 0$. Consequently $h_\infty = e^{-k\omega\cdot x}$, up to its known sign, and the same $W = c_\infty$ follows. Finite-height escaping sequences reduce to shifted plane limits. Negative heights follow by reflection.

The unique bounded translate limit implies vanishing local form norm of $V - c_\infty$ at infinity. For the bounded part this uses compactness of the set of squares of a bounded H^1_0 ball in L^1, not convergence in L^∞ norm. An IMS partition then yields, for every small $\epsilon > 0$, an exterior radius with

$$q_V[u] \geq (\tfrac{1}{2} - \epsilon)\|\nabla u\|_2^2 + (c_\infty - \epsilon)\|u\|_2^2 \tag{25}$$

for exterior-supported u.

Choose $k < a < \sqrt{k^2 + 2d}$. Testing $H_V g = 0$ with a truncated exponential weight and an exterior cutoff gives $e^{a|x|}g \in H^1$. The trace theorem would then give a finite L^2 trace on the plane. But $g(p,0) = \sqrt{n(p,0)}$ and (9) imply a divergent lower bound proportional to

$$\int_{p_0}^{\infty} e^{2(a-k)p} p^{2\beta-7}\,\mathrm{d}p = \infty. \tag{26}$$

Thus $V \notin \mathcal{V}$. The strict gap $d > 0$ provides the interval for a; the argument does not apply unchanged at $d = 0$.

8 A strictly larger excluded potential class

Definition 8.1 (Vanishing local absolute form remainder)**.** A real $F \in L^1_{\mathrm{loc}}(\mathbb{R}^3)$ belongs to $\mathcal{R}$ if it is globally infinitesimally form bounded in absolute value,

$$\int |F||u|^2 \leq \epsilon\|\nabla u\|_2^2 + C_\epsilon\|u\|_2^2 \quad (u \in H^1), \quad \text{for every } \epsilon > 0, \tag{27}$$

and, for every finite $L > 0$,

$$a_L(y) := \sup_{\substack{u \in H^1_0(B_L) \\ \|u\|_{H^1} \leq 1}} \int_{B_L} |F(y+x)||u(x)|^2\,\mathrm{d}x \longrightarrow 0 \quad (|y| \to \infty). \tag{28}$$

Define $\mathcal{V}_{\mathrm{vf}} = \mathcal{R} + L^\infty(\mathbb{R}^3;\mathbb{R})$.

The absolute value and the uniform supremum are deliberate. Weak cancellation of signed potentials would not supply the Caccioppoli absorption or mixed-form limit used below. This class consists of multiplication potentials, not all distributions that act as Sobolev forms.

Lemma 8.2. *$\mathcal{V} \subset \mathcal{V}_{\mathrm{vf}}$, and $\mathcal{V}_{\mathrm{vf}}$ defines closed lower-semibounded forms on H^1.*

Proof. For $F \in L^{3/2}$, split F into a bounded truncation and an $L^{3/2}$ tail of arbitrarily small norm. Sobolev's inequality gives (27). On an escaping fixed ball the $L^{3/2}$ norm tends to zero, again giving (28). The bounded summand is harmless. The infinitesimal form perturbation theorem proves closedness and lower semiboundedness. □

Theorem 8.3 (Extended conditional nonrepresentation)**.** *Under the fixed-density package in Section 7, no real spin-independent potential in $\mathcal{V}_{\mathrm{vf}}$ represents n as a three-electron noninteracting ground-state ensemble. In particular the package forces $V \notin \mathcal{V}_{\mathrm{vf}}$.*

Proof. First assume, for a contradiction, that $V = F + B \in \mathcal{V}_{\mathrm{vf}}$. The formal V is even in z. Averaging both summands with their reflections preserves the absolute form assumptions and boundedness, so take F, B even as well. Along any escaping centres, extract a weak-* limit W of $B(y_j + \cdot)$ on all fixed balls by a diagonal argument. Normalized orbitals have local L^2 bounds because their squares are dominated by the normalized density. Testing their equations against a compact cutoff squared times the orbital, the negative part of F is absorbed by (28); bounded B, the fixed eigenvalues, and the cutoff derivative give local H^1 bounds. No L^p norm of F is needed at this step.

For any fixed compact test function φ and a locally H^1-bounded orbital sequence u_j, polarization through the positive measure $|F(y_j + x)|\,\mathrm{d}x$ gives

$$\left|\int F(y_j + x) u_j \varphi \,\mathrm{d}x\right| \leq \left(\int |F(y_j + x)||\chi u_j|^2\right)^{1/2} \left(\int |F(y_j + x)||\varphi|^2\right)^{1/2} \longrightarrow 0,$$

where $\chi = 1$ on the test support. Rellich compactness handles the bounded part with strong local L^2 convergence. Thus the limiting orbital equations have bounded potential W, exactly as in Section 7. The plane Harnack argument and the subsequent half-space truncation test first establish the needed angle separation. The bulk angle equation and finite-height reduction then identify every such limit as $c_\infty = d + k^2/2$.

For a fixed ball, this identification gives

$$\sup_{\|u\|_{H^1_0(B_L)} \leq 1} \left|\int_{B_L} (V(y + x) - c_\infty)|u|^2 \,\mathrm{d}x\right| \longrightarrow 0. \tag{29}$$

Indeed the F part is bounded by $a_L(y)$. If the assertion failed for the bounded part, choose offending y_j, u_j. The u_j have a strongly L^2-convergent subsequence, hence $|u_j|^2$ converges in L^1. The uniform L^∞ bound and weak-* convergence to c_∞ then contradict the offending inequality.

Use a fixed locally finite quadratic partition with bounded overlap and bounded $\sum |\nabla \chi_\nu|^2$. Apply (29) to $\chi_\nu u$ outside a sufficiently large radius. Summing yields

$$\left|\int (V - c_\infty)|u|^2\right| \leq \delta\{\|\nabla u\|_2^2 + C\|u\|_2^2\}.$$

Taking δ suitably small gives (25). Since $V \in \mathcal{V}_{\mathrm{vf}}$ under the contradiction hypothesis, its form is now legitimately closed on H^1, and g is a form eigenfunction. Let $f_T = a \min(\rho, T)$ with a smooth radial ρ agreeing with $|x|$ outside a compact set and $|\nabla \rho| \leq 1$. For an exterior cutoff η, the eigenfunction identity is

$$q_V[\eta e^{f_T} g] = \tfrac{1}{2} \int g^2 |\nabla(\eta e^{f_T})|^2.$$

Choose a small cross-term parameter $t > 0$ so that $(1 + t)a^2/2 < c_\infty - \epsilon$. The right side is at most $(1 + t)a^2 \|\eta e^{f_T} g\|_2^2/2 + C_t$, where C_t is supported on a fixed annulus and independent of T. Inequality (25) now bounds both the L^2 norm and gradient uniformly. Letting $T \to \infty$ gives $e^{a|x|} g \in H^1$ outside, hence globally after a harmless compact modification. Its plane trace contradicts (26).

Finally suppose a representing $v \in \mathcal{V}_{\mathrm{vf}}$ exists. Its absolute form bound gives finite kinetic energy of a ground-state ensemble by the many-particle lower bound $H_s(v) \geq cT - C$ in form sense. The source's constrained-search identity (23) therefore again makes Φ a ground state. The down-spin form variation gives $v = V + e_g$ distributionally and almost everywhere. The products are locally integrable, and g is locally positive. Closure under addition of constants then forces $V \in \mathcal{V}_{\mathrm{vf}}$, the contradiction just proved. □

Proposition 8.4 (Strictness of the extension)**.** *Let* $x_j = 4je_1$, $r_j = \frac{1}{10}j^{-2/3}$, *and*

$$F(x) = \sum_{j=1}^{\infty} j\,\mathbf{1}_{B(x_j,r_j)}(x). \tag{30}$$

Then $F \in \mathcal{R}$, *but* $F \notin L^{3/2} + L^{\infty}$.

Proof. The balls are separated and the sum is locally finite. The $L^{3/2}$ norm of the jth spike, rather than its $3/2$ power, is proportional to

$$jr_j^2 = 10^{-2}j^{-1/3} \longrightarrow 0.$$

Local Sobolev bounds on disjoint fixed-radius neighbourhoods, with cutoff derivative terms, yield

$$\sum_{j>J}\int_{B(x_j,r_j)} j|u|^2 \le C\sup_{j>J}(jr_j^2)(\|\nabla u\|_2^2 + \|u\|_2^2).$$

The finitely many remaining spikes are bounded by $J\|u\|_2^2$. This proves (27). A fixed-radius escaping ball intersects at most a number of these neighbourhoods depending on its radius; every intersected index tends to infinity. The same estimate proves (28).

If $F = A + B$ with $\|B\|_\infty = M$, then on every spike with $j > 2M$ one has $|A| \ge j/2$. Therefore

$$\int |A|^{3/2} \ge C\sum_{j>2M} j^{3/2}r_j^3 = C'\sum_{j>2M} j^{-1/2} = \infty,$$

contradicting $A \in L^{3/2}$. This excludes every bounded subtraction, not just the decomposition $B = 0$. □

The strictness example is an analytic potential-class example. It is not a proposed Coulomb nuclear potential or a computed Kohn–Sham inversion. The theorem extends the fixed-density exclusion; it does not relax the source's positive gap, tail, or angle hypotheses.

9 Charge selection and what remains nonnumerical

The source fixes $D = 10^{200}$ and uses a tolerance of 10^{-60} in several uniform exterior estimates. It selects an integer exterior scale $M > 10D$ using limits and limsups, then $S = 100M$ and a positive inner-cylinder margin. A coupling is declared bad if it fails regularity, exterior tests, or one of the required inequalities for any folded minimizer. Its charge is of the form

$$Z = 2 + \left\lceil \sup_{0<t\le 1} t^{-1}\mathbf{1}_{\{t \text{ bad}\}} \right\rceil.$$

The proof that a full small interval is good makes this supremum finite and implies $1/Z$ is not bad. This is an exact existence specification, but not an evaluated integer, finite computation, or practically accessible atom. A finite set of sampled couplings cannot replace the universal condition over a punctured interval and all minimizers.

A second logical distinction matters. The final good coupling selected this way need not be explicitly known to lie in a previously named perturbative interval. The source therefore supplies a fixed-parameter channel proposition with its own hypotheses. It must be invoked at that stage,

rather than extending small-coupling estimates by informal continuity. The reported inequalities $2\epsilon_b - E_3 > 0.49$ and $0.99 < k < \gamma$ participate in this fixed-parameter argument.

Ram residual and gap certificates could help replace parts of this construction by interval bounds, but the current report does not supply certified M, Z, Coulomb eigenfunctions, or angular amplitudes. Conditional formulas with an unknown positive input are not numerical certificates for that input.

10 Functional consequences and limits of differentiability claims

In the usual density Banach space $X = L^1 \cap L^3$, the dual potential space is $X^* = L^\infty + L^{3/2}$. Let F and T_s be the interacting and noninteracting ensemble constrained-search functionals on their common finite-energy density domain. A subgradient $u \in \partial_{X^*} T_s[n]$ would make an attaining kinetic minimizer a ground state for $-u$. Consequently the source's nonrepresentation result implies

$$\partial_{X^*} T_s[n] = \varnothing.$$

This conclusion uses both attainment at n and the ensemble variational argument.

Let $E_{\mathrm{Hxc}} = F - T_s$. Suppose a single $w \in X^*$ represented its one-sided derivatives along *all* admissible density segments $n_t = n + t(m-n)$. The interacting support inequality with external potential v_{ext}, combined with convexity of T_s, gives

$$T_s[m] - T_s[n] \geq \frac{T_s[n_t] - T_s[n]}{t} \geq -\int v_{\mathrm{ext}}(m-n) - \frac{E_{\mathrm{Hxc}}[n_t] - E_{\mathrm{Hxc}}[n]}{t}.$$

Taking the assumed limit yields a subgradient $-v_{\mathrm{ext}} - w$, a contradiction. This excludes that simultaneous dual representation. It does not exclude directional derivatives on a smaller set, regularized functionals, finite-basis derivatives, or generalized potential spaces.

The larger class $\mathcal{V}_{\mathrm{vf}}$ is not asserted to be the Banach dual of $L^1 \cap L^3$. Theorem 8.3 is an operator/form statement. Transporting the functional conclusion to another topology would require defining its density space, duality, lower semicontinuity, and segment domain anew.

11 Current literature and noncontradictions

The earlier nodal-plane analysis of Gori-Giorgi, Gál and Baerends [6] motivates the source's tail mechanism: unusual density decay in a nodal direction can demand irregular scalar potentials. The later analysis of Gori-Giorgi and Baerends [7] studies the effective equation for the square root of the density. That effective potential must not be identified without proof with one scalar potential supporting all occupied Kohn–Sham orbitals. Here the constrained angle and the Pauli argument supply precisely the missing common-orbital structure.

Corso's 2026 bounded-interval result [8] concerns a one-dimensional setting with an enlarged distributional potential class and specified boundary realization. It is not a three-dimensional existence theorem in (1), and so does not contradict the source's proposed example. Corso's more recent uniqueness result [9] concerns the relationship between density positivity and potential uniqueness in a broad form-bounded setting. Uniqueness conditional on representation does not prove that a representing potential exists. The statements and hypotheses, not a complete independent proof audit of these papers, were checked for this comparison.

Numerical atomic inversions that fail an Aufbau ordering [10] address a related warning about representation. Numerical evidence or failure of a particular inversion is not by itself a proof of

absence of every ground-state ensemble. Conversely, the present exact obstruction does not measure the practical error of a finite approximate calculation.

The public literature check was conducted on October 8, 2026. The present contribution is not claimed to have established historical priority for each weighted-form or spectral estimate. Its source-specific gain is the rigorously delimited synthesis and the larger excluded remainder class under an explicit fixed-density package.

12 Implementation and reproducibility plan

1. Fix a density and an admissible angle space, parity, normalization, and exterior boundary treatment. Track $\int n \sin^2\theta - 1$ separately from the differential defect.

2. Assemble G, K, L with physically weighted integrals. Certify quadrature and tail errors before interpreting interval eigenvalue bounds. Confirm linear independence through a lower bound on G.

3. Establish an analytic complementary bound; combine it with the full residual coupling using Theorem 5.2. Record $\alpha - d$, $\beta - d$, and the determinant margin, not only rounded eigenvalues.

4. Measure the angle defect with $\int |r|^2/n$. Report density error with a potential-weighted norm when using Proposition 6.1.

5. Treat ionic gap, multipole remainder, fast-channel crossover, angular slope, and fixed-parameter tail conditions as independent inputs requiring proof or interval bounds.

6. For nonrepresentation, prove the universal escaping local form condition, not a finite-radius numerical observation. Preserve the strict positive interval $k < a < \sqrt{k^2 + 2d}$.

The accompanying exact-arithmetic and symbolic checks concern the product-rule residual, the 2×2 certificate, Pauli bookkeeping, the spike exponents, and strict decay thresholds. They do not simulate the constructed molecule, certify the source's nonnumerical charge, or establish the full theorem by finite sampling.

13 Advantages, limitations, and future work

13.1 Advantages over an unstructured reformulation

The weighted realization makes the Hilbert space and domain explicit even when the inverted potential is outside a conventional class. Physical Gram matrices preserve Rayleigh quotients and expose errors due to coefficient conditioning. The finite/complement certificate states exactly what finite computations still need from analysis. The angle and density defect identities quantify where apparently small residuals can be misleading. The extended remainder class replaces global integrability by the local property actually used at infinity, while retaining a closed-form realization. These are useful analytic and computational controls, not merely new notation for the original formulas.

13.2 Costs and boundaries

The transformed weight can be extremely small and anisotropic, making discretization difficult. The complement estimate and source channel asymptotics remain demanding. The class extension

retains a uniform absolute form condition; it is not an exclusion of arbitrary singular or distributional potentials. No universal speedup, superior numerical precision, experimentally relevant counterexample, or efficient evaluation of Z is established. The source's delicate construction is not made elementary by a coordinate transform. A useful conditional extension is distinct from an independently verified new unconditional Coulomb theorem.

13.3 Research directions

An interval-certified moderate-scale example would be a major advance beyond the current nonnumerical parameter choice. It would require certified two-electron and three-electron energies, a tail normalization with nonzero angular slope, and uniform minimizer bounds. A second direction is an a posteriori angle existence theorem that uses Proposition 3.1 together with an invertible constrained linearization; residual smallness alone does not give such a theorem. Third, one can investigate codimension and nodal-order variants of the trace contradiction, with the appropriate Sobolev trace threshold proved rather than guessed. Fourth, weighted frames adapted to the three scales $A|\omega_z|$, b, and $\lambda A r^{-4}$ may improve conditioning, but need physical-metric stability and tail quadrature analysis. Finally, distributional potentials outside $\mathcal{V}_{\mathrm{vf}}$ require a new domain and positivity analysis, in dialogue with the current uniqueness literature.

14 Assessment and conclusion

Applicability is L4: the core obstruction is a continuum many-body and asymptotic argument, while Ram physical metrics, domain transport, residual accounting and form localization are directly useful in significant parts. Contribution is C3, independently of applicability. The present report proves a conditional enlargement of the excluded scalar-potential class, including a strict example, and several quantitative certificates with explicit limitations. It does not close an identified gap in the source's entire Coulomb proof, verify all of its difficult uniform estimates, or supply an evaluated nuclear charge. The strongest defensible conclusion is an improved fixed-density obstruction framework whose hypotheses and computational obligations are visible and testable.

References

[1] OpenAI mathematics repository, *A Coulomb ground-state density without Kohn–Sham ensemble representation*, September 25, 2026. Pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, preprint build, main text and sections 1–7. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-Coulomb-Ground-State-Density-without-Kohn-Sham-Ensemble-Representation-September-25-2026.

[2] M. Subbarao, Ram Calculus books, volume 21, physical-metrics chapter, local source `volumes/volume21/chapters/ch01d-physical-metrics.tex`. Entire chapter consulted; private supporting manuscript.

[3] M. Subbarao, Ram Calculus books, volume 21, state/domain chapter, local source `ch01a-state-domain.tex`, lines 71–170 consulted. Private supporting manuscript.

[4] M. Subbarao, Ram Calculus books, volume 21, spectral-limits chapter, local source `ch01f-spectral-limits.tex`. Entire chapter consulted; private supporting manuscript.

[5] M. Subbarao, Ram nonlinear integral-equation/PDE corpus, `TR8-lin-non-lin/TR8-lin-non-lin.tex`, lines 61–201, PDE-first domains and form realizations. Private supporting manuscript.

[6] P. Gori-Giorgi, T. Gál and E. J. Baerends, *Asymptotic behaviour of the electron density and the Kohn–Sham potential in case of a Kohn–Sham HOMO nodal plane*, arXiv:1109.2204, version 2. https://arxiv.org/abs/1109.2204.

[7] P. Gori-Giorgi and E. J. Baerends, *Asymptotic nodal planes in the electron density and the potential in the effective equation for the square root of the density*, arXiv:1804.00475 (2018). https://arxiv.org/abs/1804.00475.

[8] T. C. Corso, *A rigorous formulation of density functional theory for spinless fermions in one dimension*, *Letters in Mathematical Physics* 116, 27 (2026), published March 3, 2026. https://doi.org/10.1007/s11005-026-02051-1.

[9] T. C. Corso, *A maximal Hohenberg–Kohn theorem for non-interacting systems via potential theory*, arXiv:2607.12852, version 3, August 4, 2026. https://arxiv.org/abs/2607.12852v3.

[10] E. Trushin, J. Erhard and A. Görling, *Violations of the v-representability condition underlying Kohn–Sham density-functional theory*, *Physical Review A* 110, L020802 (2024). https://doi.org/10.1103/PhysRevA.110.L020802. Publisher abstract consulted.